How Sealroom works

Two law firms find out whether they share clients without either one showing its client list to anyone. Then the firm's AI assistant works with the conflicted clients walled off. Every step leaves a receipt anyone can check.

Built on NEAR + NEAR AI Cloud
One-page overview, November 2026

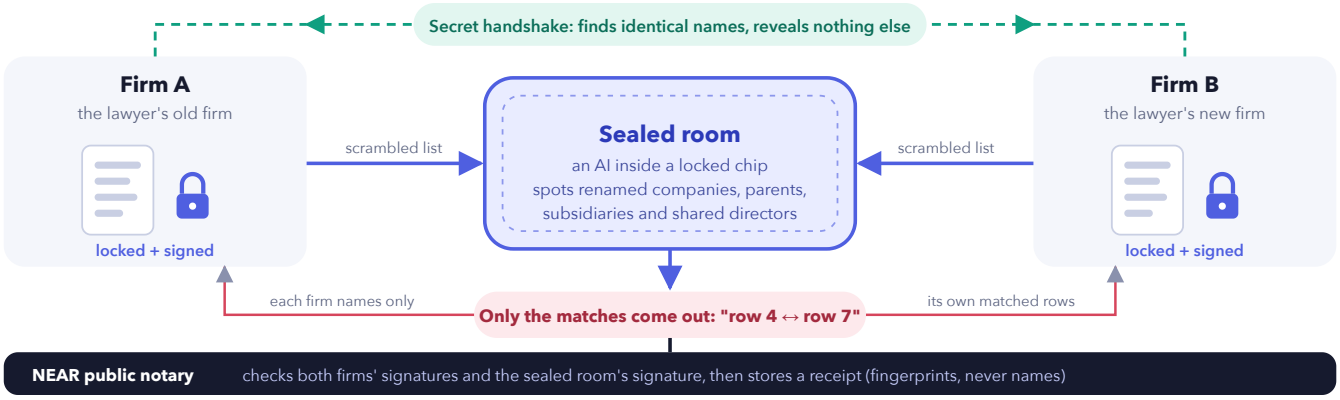
The problem

When a lawyer moves firms, both firms must check for **conflicts of interest**: is the lawyer's old client someone the new firm is suing, or acting for? Today one side hands its **confidential client list** to the other, or to a vendor, and simply has to trust them.

The idea

Each firm locks its list in its own browser and drops it into a **sealed room**: a computer chip that even its operator cannot look inside. Only the matches come out. A **public notary** (the NEAR blockchain) checks everyone's signatures and keeps a receipt.

1 The conflict check



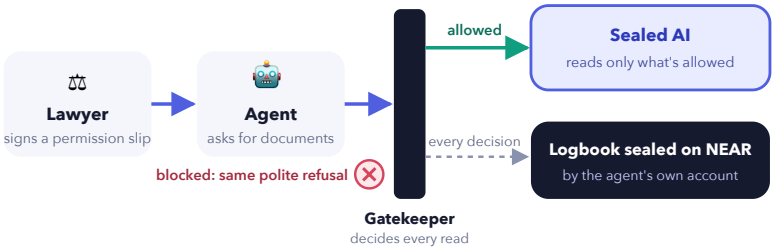
1. **Check the room is genuine**
   Intel's certificate proves the chip is real, sealed and unmodified.
2. **Lock and sign**
   Each firm scrambles its list in its own browser and signs it with its registered key.
3. **Secret handshake**
   Exact name matches ("Vale Doce S.A." = "Vale Doce") are found by maths alone.
4. **The sealed AI looks deeper**
   It answers with row numbers only, never names, so nothing else can leak.
5. **The notary records it**
   NEAR refuses the record unless every signature checks out.
6. **Ranked result**
   "Old client vs. the new firm's opponent" comes first, then cases to review.

2 Everyone has an ID card

🏛 **Firm**	Registers its own signing key. A list it didn't sign is rejected.
⚖ **Lawyer**	Own NEAR account, added by the firm. Signs every AI session they start.
🤖 **AI agent**	Owned by a partner, locked to one AI model and one set of instructions. Its key can do exactly one thing.

Like a building pass, with NEAR as the front desk that checks it every time. A firm can cancel a pass at any moment.

3 The AI assistant, with walls



Conflicted clients from the check become walls. Every blocked read gets the same refusal, so the AI can't work out who the firm acts for.

4 Why you don't have to trust us

- ✓ No list ever leaves a firm unscrambled. Our server only sees row numbers.
- ✓ Each firm can later prove exactly which list it submitted, without showing it.
- ✓ Tampering with the result, the AI's instructions or a reply is detected.
- ✓ Every result comes with a **proof pack** anyone can replay to get the same answer.
- ✓ The notary rejects anything with a missing or forged signature.
- ✓ The AI agent can't swap its model or instructions, or act without a lawyer's signature.

Sealed room (enclave)
A sealed processor whose memory is encrypted. Intel signs a certificate saying exactly what runs inside.

NEAR
A public blockchain. Here it is a notary: it stores receipts and checks signatures, but never sees client names.

Signature
A digital stamp only the key's owner can make, and anyone can check against their registered ID.

Fingerprint
A short code computed from a list. It proves which list was used but can't be turned back into the list.